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EXPERIMENT: OSTWALD METHOD VISCOSITY

Objectives:

1. Determining the viscosity of a solution using the Ostwald method.

EXPERIMENT SUMMARY

INTRODUCTION

Water transferred from a bucket to a new container will conform to the shape of its new container, just as soy sauce or honey conforms to the shape of its storage container. This is because water, soy sauce, and honey are fluids. A fluid is a substance that can flow and has the ability to move and flow when subjected to a force, and can adjust to the shape of its container, meaning it has no fixed shape. Because of its ability to move when subjected to a force, a fluid has a property of viscosity, or resistance to flow, which is what causes water and honey to have different degrees of difficulty in flowing. This degree of difficulty in a fluid's flow is referred to as viscosity [1].

Viscosity is a fluid property that describes how much internal resistance the fluid has to flow or deformation. Simply put, viscosity can be understood as the "thickness" of a fluid, where the higher the viscosity, the greater the internal frictional force between layers of fluid as it flows [2].

In this practicum, three fluids were used as test materials for viscosity measurement using the Ostwald method, namely distilled water, 60% v/v ethanol solution, and 70% v/v ethanol solution. Data collection consisted of the time required for each fluid to reach a certain point using an Ostwald viscometer tube, as shown in Figure 1. The Ostwald viscometer is an instrument used to determine the viscosity of a liquid based on the principle of fluid flow through a capillary tube under the influence of gravity. Its basic principle relies on the Poiseuille equation, which states that the flow rate of a fluid through a capillary tube is inversely proportional to its viscosity [3].

The equation used to obtain the viscosity value of the fluids used is the Hagen-Poiseuille equation. The Hagen-Poiseuille equation describes the relationship between the flow rate of a viscous fluid flowing laminarly through a cylindrical (capillary) tube and various factors such as pressure, tube radius, tube length, and the fluid's own viscosity. This equation was first formulated separately by Gotthilf Hagen and Jean Léonard Marie Poiseuille in the mid-19th century, and it forms the primary theoretical basis for viscosity measurement using capillary methods, including the Ostwald viscometer. The equation is written mathematically as follows :

$$\frac{v}{t} = \frac{\pi r^4 \Delta P}{8\eta L} \tag{1}$$

where r is the radius of the capillary tube (m), ΔP is the pressure difference between the two ends of the tube (Pa), η is the dynamic viscosity of the fluid (Pa·s), L is the length of the capillary tube (m), v is the volume of fluid (m^3), and t is the flow time (s) [4].

In the Ostwald viscometer, the fluid flows not as a result of external pressure but due to gravitational force, so ΔP is replaced by the hydrostatic pressure of the liquid itself, namely ρgh, where ρ is the density of the liquid (kg/m^3), g is the acceleration due to gravity (m/s^2), and h is the difference in fluid height (m). Thus, by substituting the value of ΔP into Equation 1, the equation for obtaining the viscosity value of the fluid is obtained as follows:

$$\eta = \frac{\pi r^4 \rho g h t}{8VL} \tag{2}$$

Since all the geometric parameters of the capillary (r, L, V, h) are constant for the same instrument, the equation can be simplified, and the viscosity value is calculated by comparing it to a standard liquid, namely distilled water, using the following equation:

$$\eta_{solution} = \eta_{distilled\ water} \times \frac{\rho_{solution} t_{solution}}{\rho_{distilled\ water} t_{distilled\ water}} \tag{3}$$

Data collection was carried out with 5 repetitions, followed by calculation of standard deviation and uncertainty to obtain the viscosity value of the solution.

METHODOLOGY

A. Tools and Materials

Listed below are the tools and materials used in the experiment:

Table 1. Tools and materials for the coefficient of friction experiment

Apparatus	Quantity	Function
Ostwald viscometer tube	1 unit	Used to measure the flow time of a liquid to determine its viscosity value.
Beaker glass	1 unit	Used as a container to hold or mix the solution to be tested.
Pipette	1 unit	Used to take and transfer a specific volume of solution into the viscometer or other equipment.
Pipette pump	1 unit	Used to help draw up and dispense solution through the pipette.
Stopwatch	1 unit	Used to measure the time required for the liquid to flow between two marks on the Ostwald viscometer.
Digital balance	1 unit	Used to measure the mass of the substance or solution.
Pycnometer	1 unit	Used to determine the density of the solution, which is needed for viscosity calculations.
Stand (statif)	2 sets	Used as a support/stand for the Ostwald viscometer tube.
Small container	1 unit	Used as a storage and waste container for the solutions used.
Distilled water (akuades)	As needed	Used as the reference (standard) liquid and for cleaning the equipment.
Ethanol solution (60% v/v and 70% v/v)	As needed	Used as test materials.
Tissue and cotton	As needed	Used to clean up spilled water and ethanol solution.

B. Procedure

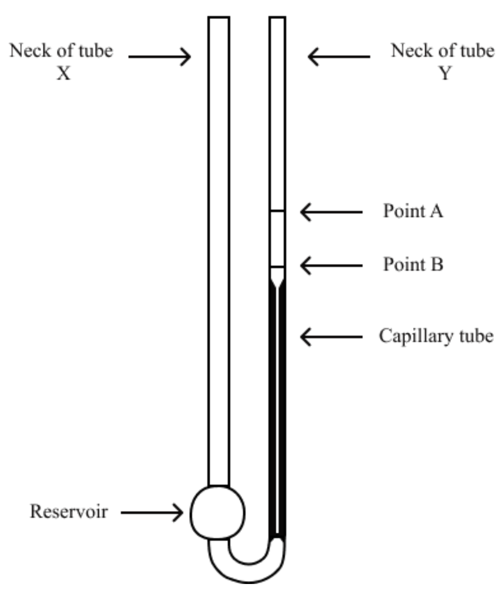

Figure 1. Schematic diagram of the Ostwald viscometer tube

The room temperature was measured using a thermometer.

1. The mass of the pycnometer along with its cap was measured using a digital balance.
2. Distilled water was poured into the pycnometer until it reached halfway up the pycnometer's neck.
3. The pycnometer was capped, and any water that overflowed from the pycnometer was dried off with tissue.
4. The mass of the pycnometer filled with distilled water was weighed, after which the distilled

water was returned to the beaker glass and the pycnometer was dried.

5. Distilled water was poured into the Ostwald viscometer tube through neck X (as shown in Figure 1) until it reached a certain height.
6. Water was drawn up using the pipette pump through neck Y (as shown in Figure 1) until it was slightly above line A.
7. The pipette pump was removed, and the time required for the water to drop from point A to point B was measured and recorded.
8. Steps 7 and 8 were repeated for a total of 5 repetitions.
9. Steps 3 through 9 were repeated for the 60% v/v and 70% v/v ethanol solutions.

C. Experiment Flowchart

DATA ANALYSIS AND DISCUSSIONS

A. Data Analysis

The following are the tables of measurement results obtained in the laboratory:

Table 2. Mass measurement results for the Ostwald method viscosity

V_{pycno}(mL)	Mass (g)			
	m_{pycno}	$m_{pycno+distilled\ water}$	$m_{pycno+60\%\ ethanol)}$	$m_{pycno+70\%\ ethanol)}$
25.0				

Table 3. Time measurement results for the Ostwald method viscosity

Repetition No.	Time (t)		
	$t_{distilled\ water}$	$t_{60\%\ ethanol}$	$t_{70\%\ ethanol}$
1			
2			
3			
4			
5			

Approved by:

Laboratory Assistant

B. Calculation Analysis

(Calculate the viscosity value using the Ostwald method with Equation 3)
The following is a sample calculation based on experimental data.
Sample Calculation:

The final calculation results for all data obtained are as follows.

Table 4. Viscosity calculation results

Time (s)			Viscosity (mPa.s)		
$t_{distilled\ water}$	$t_{60\%\ ethanol}$	$t_{70\%\ ethanol}$	$\eta_{distilled\ water}$	$\eta_{60\% ethanol}$	$\eta_{70\% ethanol}$
Average					
Standard deviation					
Uncertainty					

C. Discussions

CONCLUSIONS

From this experiment, it can be concluded that:

1.

REFERENCES

[1] D. Halliday, R. Resnick, and J. Walker, *Fundamentals of Physics Extended*, 9th ed. Hoboken, NJ: John Wiley & Sons, 2010.

[2] F. M. White, *Fluid Mechanics*, 8th ed. New York: McGraw-Hill Education, 2016.

[3] P. Atkins and J. de Paula, *Atkins' Physical Chemistry*, 10th ed. Oxford: Oxford University Press, 2014.

[4] B. R. Munson, D. F. Young, T. H. Okiishi, and W. W. Huebsch, *Fundamentals of Fluid Mechanics*, 7th ed. Hoboken, NJ: John Wiley & Sons, 2013.

PRELIMINARY ASSIGNMENT

1. What assumptions are used in the Hagen-Poiseuille law?
2. Why is data collection for all fluids carried out using the same single Ostwald viscometer tube?
3. Would the viscosity value of a fluid obtained from two different Ostwald viscometers have different magnitudes?
4. What factors influence the phenomenon of viscosity in liquids?